



Qubika

Introduction to Data Engineering

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What is Data Engineering?

Main Goals:

- Make data available, reliable, and accessible
- Build scalable data pipelines
- Support analytics, machine learning, and BI

What is Data Engineering?

Key Responsibilities:

- Ingest data from APIs, databases, files, etc.
- Transform raw data (cleaning, filtering, joining)
- Store data in data lakes, warehouses, or databases
- Automate workflows (scheduling, monitoring)
- Ensure data quality, validation, and security

Core Concepts

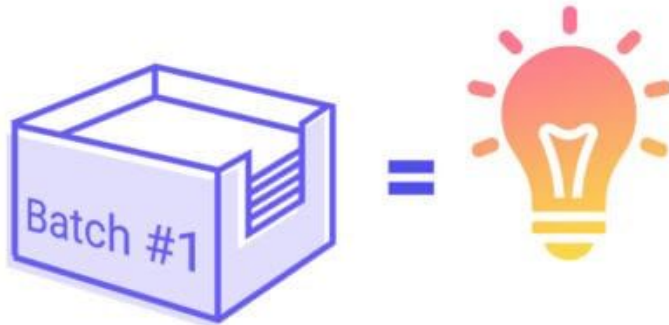
- Data Sources
- Ingestion
- Storage
- Orchestration

Data Ingestion

Manual or Scheduled Bulk Loading

BATCH

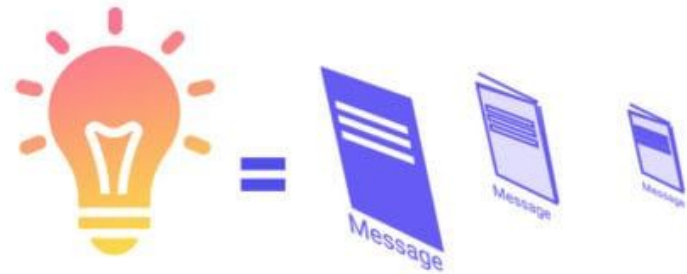
Needs all information packages to start processing



Continuous Flow of Data

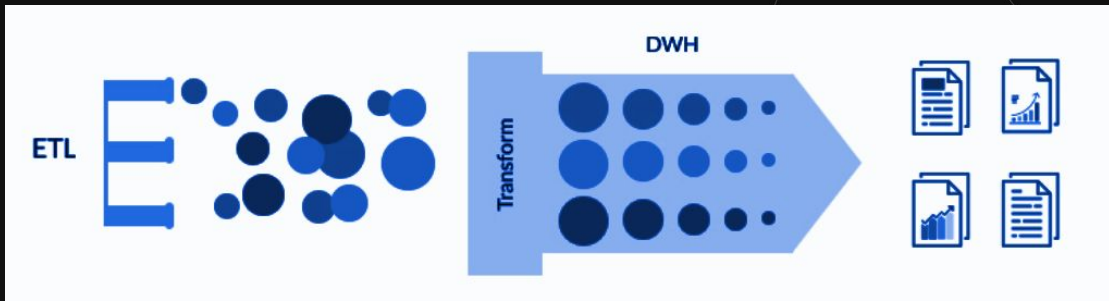
STREAMING

Needs one information package only

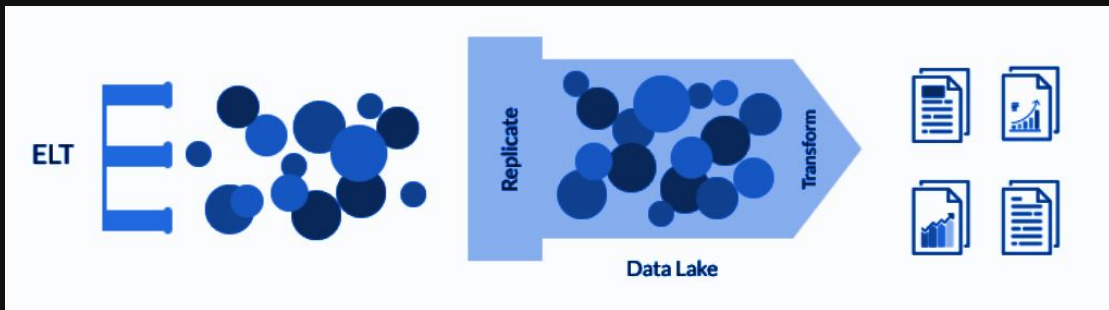


ETL vs ELT

Extract → Transform → Load



Extract → Load → Transform



The Data Storage Hierarchy

Fundamental Database

- Data Storage Form

Data Lake

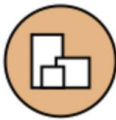
- Raw Materials for Deep Dives

Data Warehouse

- The Central Data Source

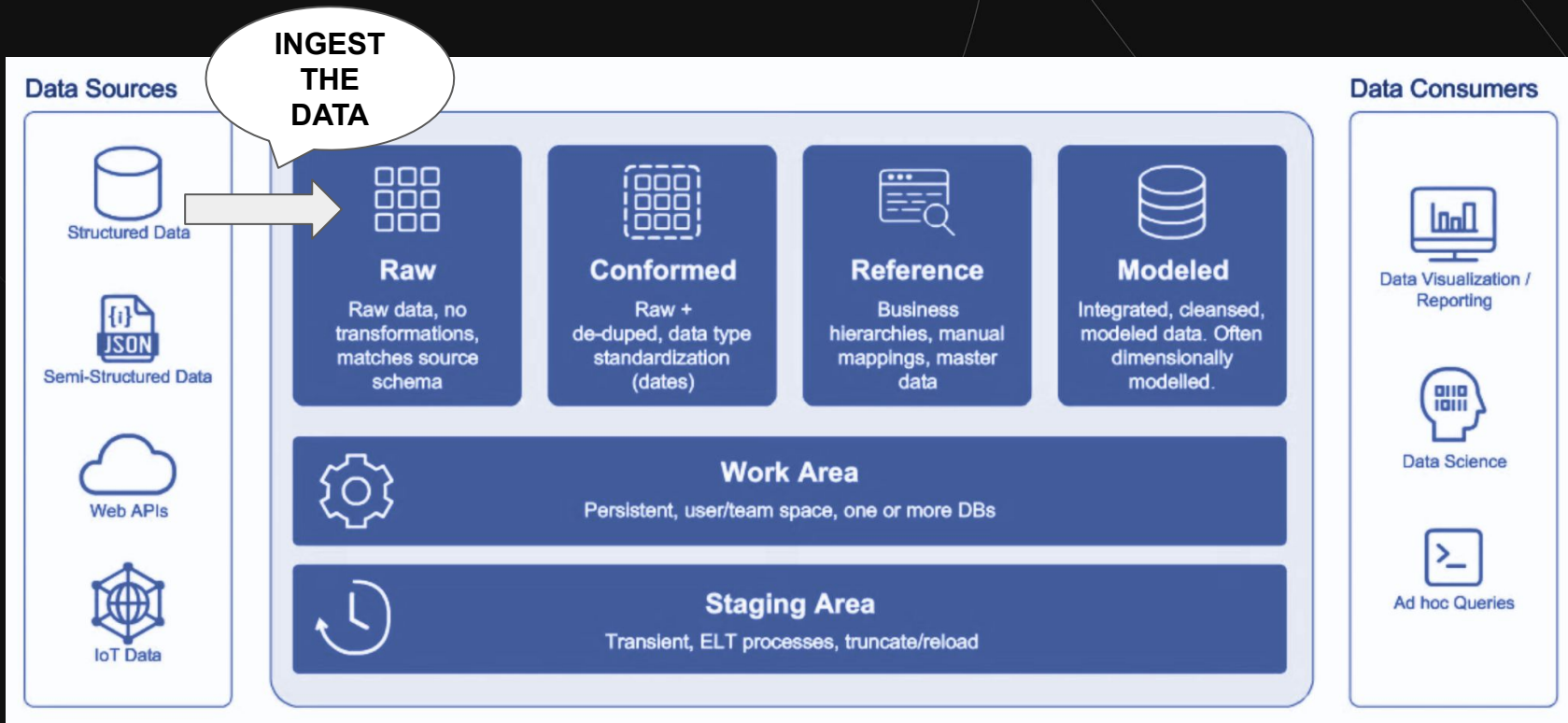
Data Mart

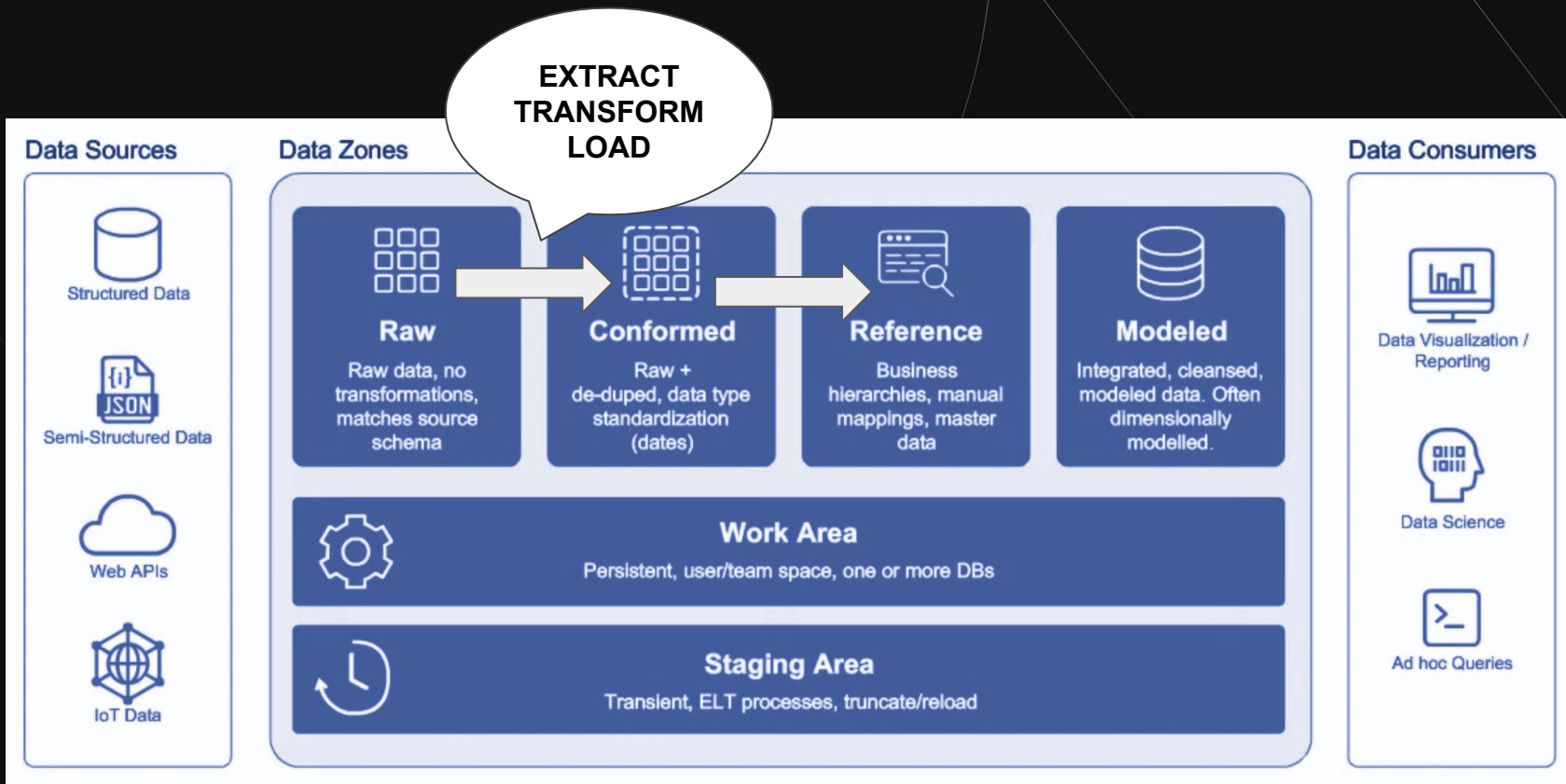
- A Focused Repository

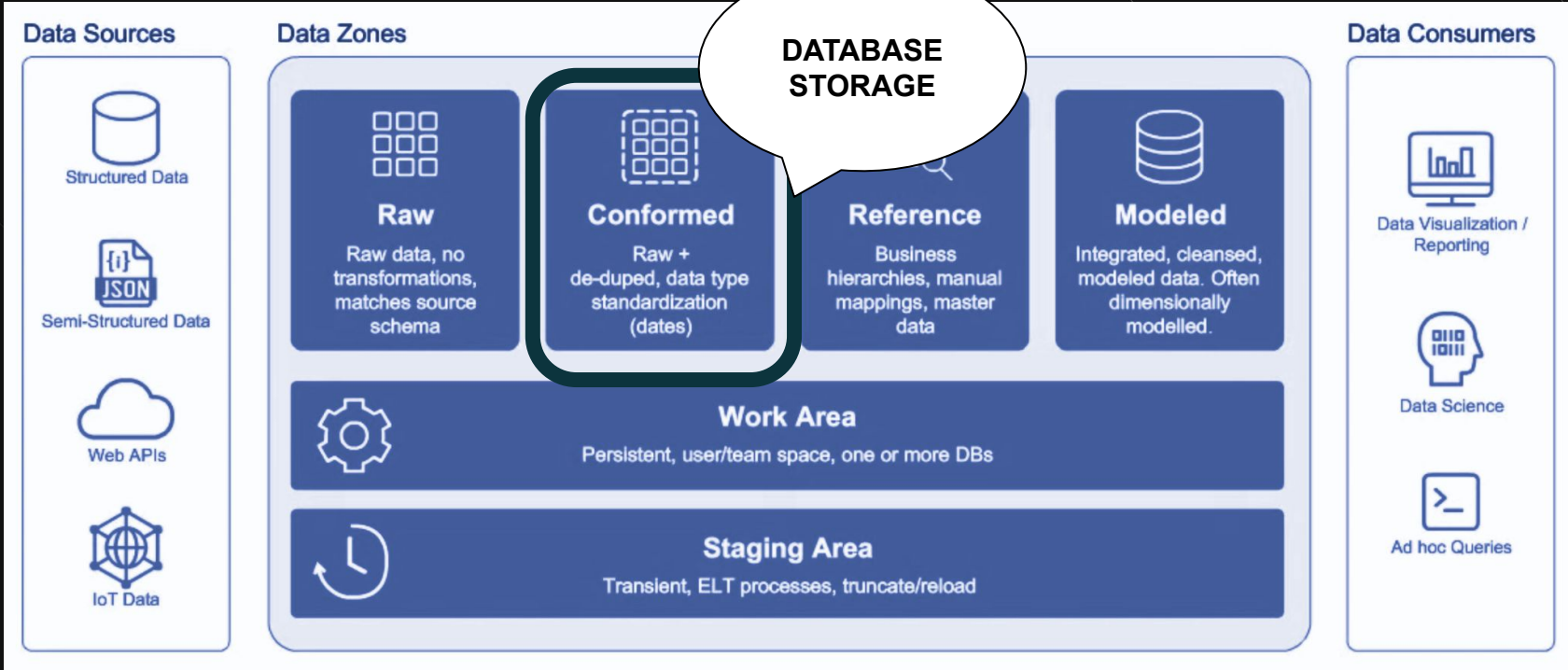
	 Data Lake	 Data Warehouse	 Data Mart
Data Scope	Broad, Raw Data	General, Cleaned	Focused
Prior Processing	None-Light	Moderate-High	Very High
Analysis Limitations	Only limited by input sources	Limited by data cleaning choices	Limited to mart topic focus
Ease of Navigation	Hard	Moderate	Easy

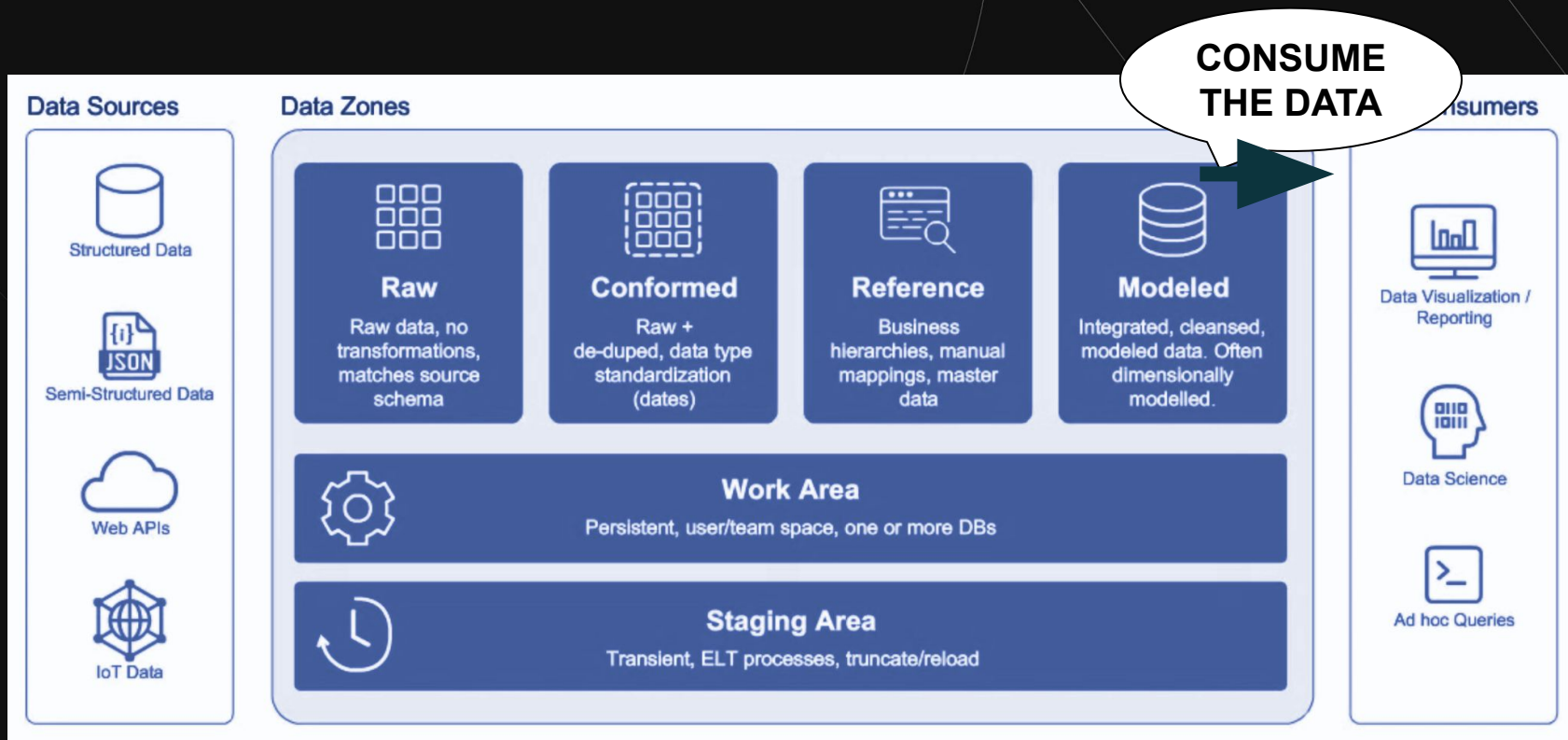
What's the big picture?

Let's decipher the warehouse architecture

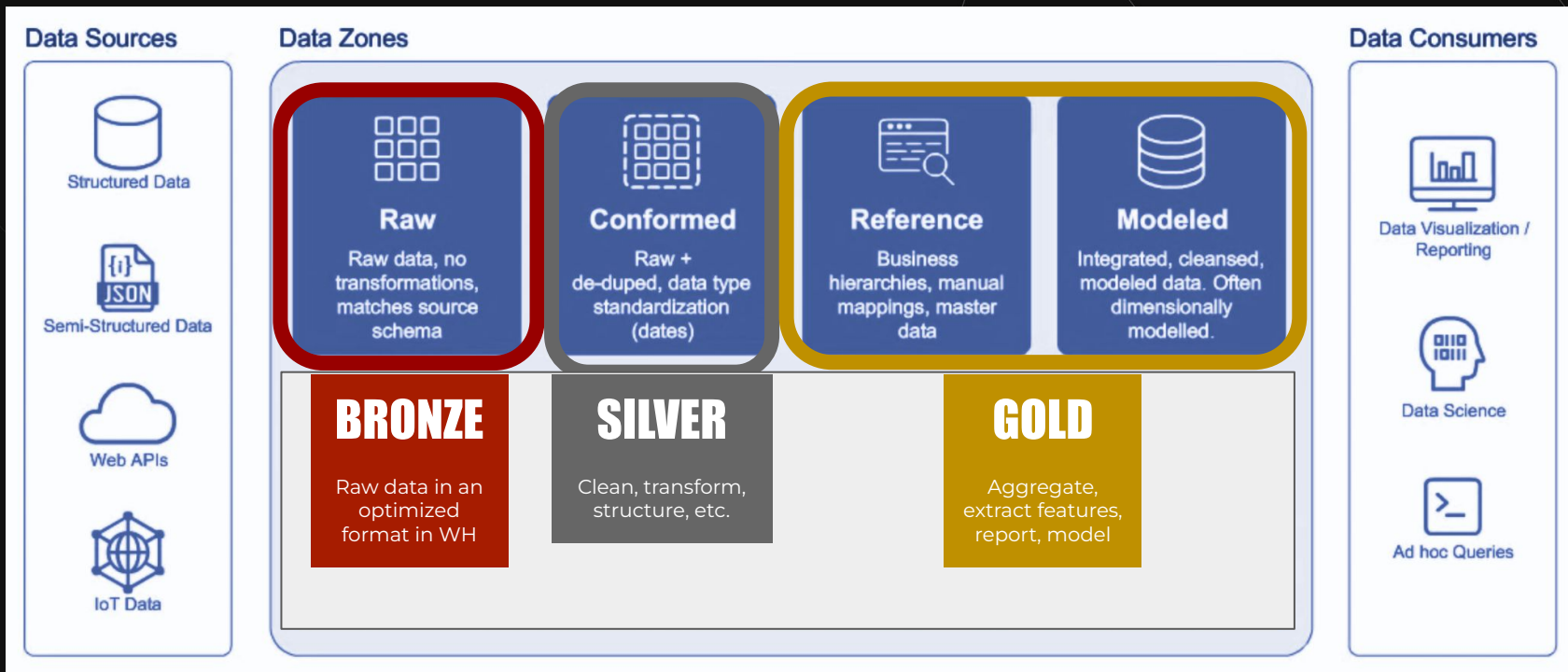








Medallion Architecture



Tools

Tool

Used For

Stage

Python

Scripting, ingestion, transformation

All stages

SQL

Querying, joining, aggregating

Silver/Gold

dbt

SQL-based transformations

Silver/Gold

Airflow

Orchestration

All stages

Power BI

Visualization & analytics

Gold

Thanks!

Qubika

We deliver excellence